

# CLAUSECHAIN

*Agentic Compliance : From Regulatory Text to Operational Action*

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Turning every SEBI circular into an **auditable, machine-actionable** compliance action — automatically.

SECURITIES MARKET HACKATHON 2026

SEBI  
ProblemStatement2—AgenticCompliance | Idea&Prototype Submission

## THE PROBLEM

# Two failures. One root cause.

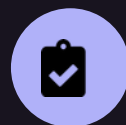
SEBI's regulatory framework moves continuously—but the systems that operationalise it are still built on manual reading, memory, and spreadsheets.



### Dynamic Regulatory Translation

Every new circular or amendment has to be read, interpreted, and manually mapped to operational processes — differently, by every intermediary, every time.

*Result: uneven, delayed, inconsistent implementation*



### Ongoing Compliance Management

Compliance teams track obligations, evidence, and audit trails largely by hand — especially at smaller intermediaries with lean compliance functions.

*Result: gaps surface only during audit or inspection*

**The root cause:** regulatory intent lives as unstructured, human-readable text, while compliance operations need structured, machine-actionable rules. Bridging that gap is what Problem Statement 2 asks for.

# The status quo doesn't scale with the rulebook

Manual interpretation was workable when circulars were rare. It isn't anymore—for regulators, large intermediaries, or the smaller firms with the thinnest compliance teams.

## Weeks

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typically spent manually reading, interpreting and re-mapping a single major circular update across teams

## Fragmented

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obligation tracking split across spreadsheets, emails and institutional memory — especially at smaller intermediaries

## After the fact

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compliance gaps are usually discovered during audit or inspection — not before they become findings

# Prose in. Rules out. Nothing bridges the two.



## Regulatory Text

- Unstructured prose
- Ambiguous, context-dependent
- Issued continuously, version over version
- Requires legal interpretation



## Compliance Systems

- Structured, queryable records
- Deterministic triggers & deadlines
- Owned by a role, tied to evidence
- Auditable at any point in time

**ClauseChain's Rule Compilation Agent is purpose-built to be this bridge** — converting cited regulatory clauses into versioned, machine-actionable Compliance Rule Objects.

# Introducing ClauseChain

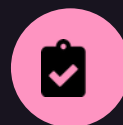
An agentic compliance platform with two engines working as one continuous loop — solving both halves of PS2 in a single, coherent system.



## Regulatory Intelligence Engine

*Solves dynamic regulatory translation*

- Watches SEBI circulars & amendments
- Extracts obligations clause-by-clause, with citations
- Compiles them into machine-actionable rule objects



## Compliance Operations Engine

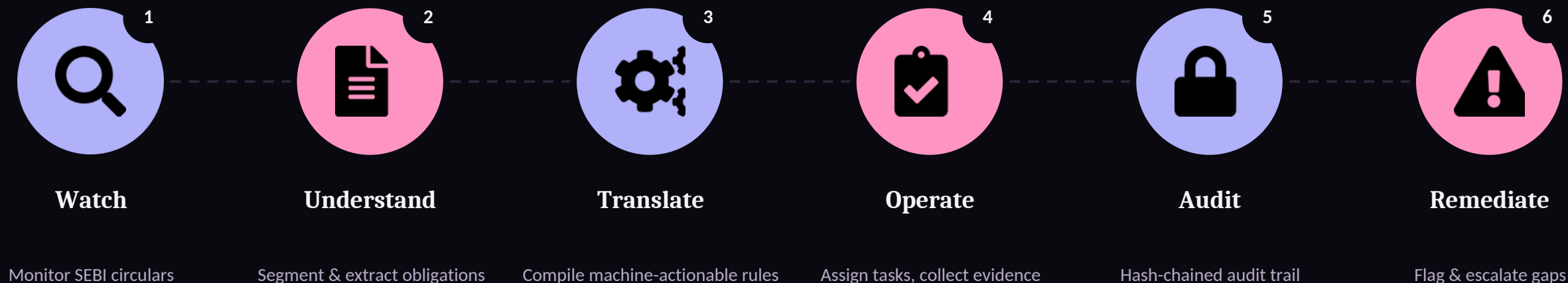
*Solves ongoing compliance management*

- Generates & assigns compliance tasks
- Links evidence to obligations automatically
- Flags gaps and maintains an immutable audit trail

*One shared knowledge base of obligations, tasks, evidence & audit history connects both engines in real time.*

## HOW IT WORKS

# A continuous loop, not a one-shot tool

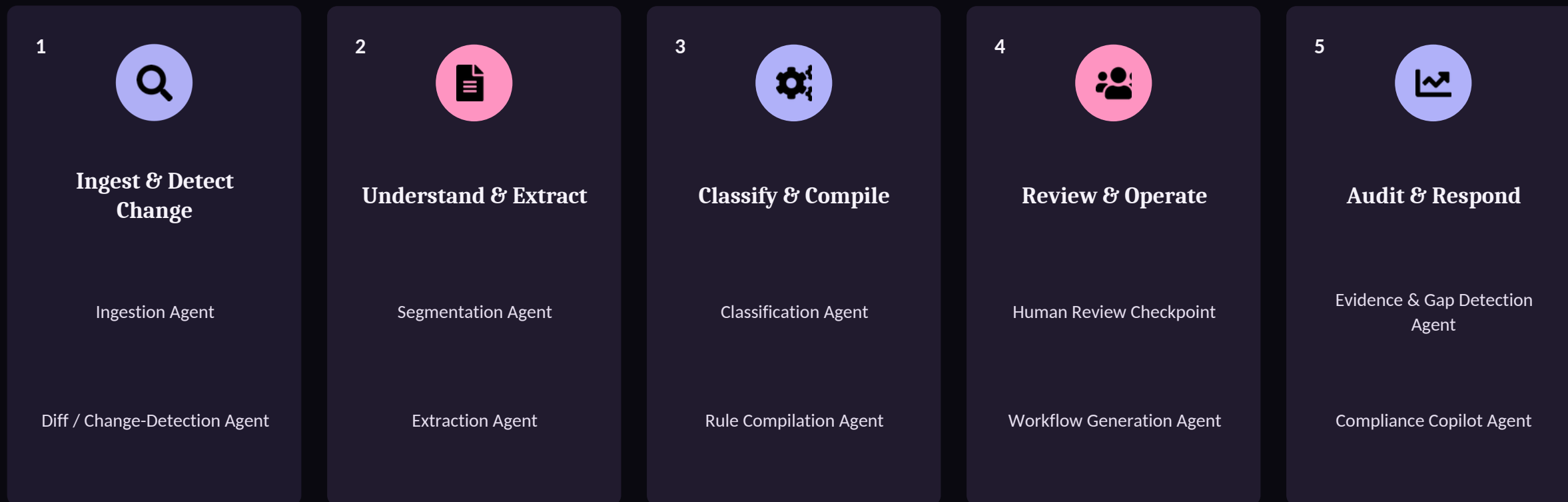


**Human checkpoints, by design:** a compliance officer approves new rule objects before they go live, and reviews evidence the system flags as borderline — the system carries the drafting and tracking load; accountability stays human.

ClauseChain • SEBI Securities Market Hackathon 2026 • PS2: Agentic Compliance

# Five stages, ten specialised agents

Each agent has one narrow job and a defined input/output—that's what makes every step traceable back to a source clause.



# Concrete scenario: Cyber Security & Resilience (CSCRF)

Chosen because it has testable parameters—clear frequency, deadlines and evidencetypes—so accuracy and correctness are objectively verifiable, live.

1

**Ingest** — Load the CSCRF section of the Master Circular for Stock Brokers

2

**Extract** — Structured, cited obligations: annual VAPT + incident reporting window

3

**Approve** — Compliance officer reviews & approves the generated rule objects

4

**Operate** — Recurring VAPT task + conditional incident-reporting task auto-created

5

**Evidence** — Sample VAPT report uploaded → task turns Compliant, audit entry logged

6

**Remediate** — A missed deadline is simulated → Gap Agent escalates automatically

*Live copilot query in the demo: “Which CSCRF obligations are currently open for our RMS/CISO team?” — answered with full clause citations.*



# Not a chatbot wrapped around a PDF

| Capability                             | ManualTeam              | Generic AI Chatbot               | ClauseChain                                                                                                                                                                                                                                                    |
|----------------------------------------|-------------------------|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Speed from circular to action          | Weeks of manual reading | Fast, but unstructured & one-off | <b>Minutes — structured, versioned, recurring</b><br><br><b>Always — clause-level citation, by design</b><br><br><b>Live compliance register + audit trail</b><br><br><b>Built-in approval checkpoints</b><br><br><b>Taxonomy-driven, multi-category ready</b> |
| Every claim cited to source            | Depends on the reader   | Rarely, and not reliably         |                                                                                                                                                                                                                                                                |
| Tracks evidence over time              | Spreadsheets & email    | No persistent state              |                                                                                                                                                                                                                                                                |
| Human accountability preserved         | Fully manual            | None — fully automated answers   |                                                                                                                                                                                                                                                                |
| Scales across obligations & categories | Breaks down at scale    | No domain structure              |                                                                                                                                                                                                                                                                |

# Production-grade architecture, not a demo hack



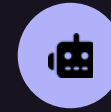
## Experience Layer

Next.js + Tailwind + shadcn/ui dashboards



## Agent Orchestration

LangGraph — stateful, resumable, retry-safe



## LLM Layer

OpenRouter — model-agnostic, RAG-grounded



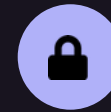
## Knowledge Store

Postgres (Neon) + pgvector, Prisma ORM



## Durable Workflows

Temporal.io for recurring compliance tasks



## Audit & Security

Hash-chained logs, RBAC, AES-256, TLS 1.3

# Targets we hold ourselves to — and demo live

≥90%

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Extraction precision on a hand-annotated clause benchmark

<1 hr

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From circular ingestion to an approved, assignable task

100%

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Compliance tasks with a fully cited, hash-verifiable evidence chain

100%

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Injected missed-deadline scenarios correctly flagged & escalated

*All four are measured against a manually annotated benchmark subset of the SEBI Master Circular for Stock Brokers — not self-reported.*

# Built for data that has legal consequences



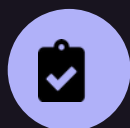
## Encryption everywhere

TLS 1.3 in transit, AES-256 at rest for evidence & audit logs



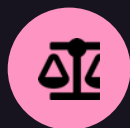
## Role-based access

Least-privilege RBAC — owner, auditor (read-only), admin roles



## Immutable audit trail

Hash-chained log entries make silent edits to history detectable



## DPDP-aligned data handling

PII in evidence documents is redacted before indexing; zero-retention LLM calls where supported

# From prototype to platform

1

**Now**

One category (stockbrokers), one demo scenario (CSCRF) — architecture proven end-to-end

2

**Next**

Extend taxonomy to Investment Advisers, DPs, RTAs & AMCs

3

**Then**

Integrate with ticketing, DMS & HR systems; explore SCORES / UIP integration

4

**Scale**

Full Temporal-based durable orchestration; open benchmark report across the full corpus

**Vision:** Every SEBI-regulated intermediary — from a two-person compliance desk to a full-scale broker — running on a live, auditable map of its obligations.



# Let's close the gap between regulation and action.

*ClauseChain — built for Problem Statement 2, ready to extend across every SEBI-regulated intermediary.*

Prototype demo & jury walkthrough

Feedback on the CSCRf benchmark  
approach

A path to pilot with a real intermediary

Team NetRunners — SEBI Securities Market Hackathon 2026